ENGINEERING CHANGE NOTE (ECN)

PROJECT: PROJECT-MYMILLI

ORIGINATION: Prof. Kidd, Instructor, UMBC

PURPOSE OF CHANGE: PWM-driven DC fan

DOCUMENT NUMBER: ECN1.3

DATE OF PUBLICATION: 10/27/2026

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TEAM LEAD: Nyi Thu Kha, UMBC

EXECUTIVE SUMMARY:

This ECN documents the change from the Project 3 (PROJECT-MYMILLI) design to add a PWM-controlled DC motor-driven fan. The customer requirements, existing high-level technical requirements, cyclic executive architecture, and custom millisecond timer are unchanged. A new pushbutton cycles the fan through OFF, LOW, MEDIUM, HIGH, MEDIUM, LOW, and back to OFF. PWM is generated on Timer2 (Arduino pin 11) using a low-side MOSFET driver with a flyback diode. A new task, taskMotor, has been added to the cyclic executive. Testing requirements have been extended to verify the new functionality. The formal design document has been revised accordingly.

HIGH-LEVEL DESCRIPTION:

Add a PWM-controlled DC motor-driven fan with a pushbutton-cycled four-speed state machine to PROJECT-MYMILLI, without changing existing customer or high-level technical requirements.

REASON FOR CHANGE:

The customer has added the new requirement of having a cooling fan added to the device.

Changed Documents:

10/27/2026 (CMPE311-PROJECT-MYMILLI-SDD1.2 → CMPE-PROJECT-ADDMOTOR-SDD.1.3) System Design Document:

- **Changed Document Name:** “Custom Millisecond Timer – System Deign Document” to “PWM Fan Controller– System Design Document”.
- **Changed Document Reference:** “CMPE311-PROJECT_MYMILLI-SDD.1.2” to “CMPE311-PROJECT-ADDMOTOR-SDD.1.3”.
- **Changed Date of Publication:** 10/15/2026 to 10/27/2026

- **HIGH-LEVEL DESCRIPTION:**
  - **Added:** “and a PWM-controlled DC motor-driven fan” at the end.
- **DESCRIPTION:**
  - **Added:** “The system also includes a DC motor-driven fan controlled by a pushbutton, with PWM generated on Timer2 pin 11 via a low-side MOSFET driver.”
- **RESULTS SUMMARY:**
  - **Added:** “and the PWM fan controller” before “were satisfied”.
- **REFERENCE:**
  - **Added:** "[1] Nyi Thu Kha, Custom Millisecond Timer - System Design Document, CMPE311-PROJECT-MYMILLI-SDD.1.2" as first reference. Previous references renumbered.
- **DEFINITIONS:**
  - **Added:** PWM, Duty Cycle, MOSFET, Flyback Diode, Debounce, CFM, RPM.
- **ACRONYSMS AND ABBREVIATIONS:**
  - **Added:** CFM, RPM, MOSFET, BJT.
- **EDUCATIONAL CONTRAINTS:**
  - **Deleted:** EC.4 (“The Evaluator must be able to successfully have The System perform the dialog in Table 5.”)
  - **Added:** EC.4 (“The Evaluator must be able to successfully have The System perform the dialog in Table 4 and Table 5.”)
  - **Added:** EC.5 (“HL.16, HL.17, HL.18 do not have to be explicitly measured. Just provide evidence via justified inference such as calculations.”)
- **PART II/III HIGH-LEVEL TECHNICAL REQUIREMENTS:**
  - **Added:** HL.7, HL.8, HL.9, HL.10, HL.11, HL.12, HL.13, HL.14, HL.15, HL.16, HL.17, HL.18.
- **ADDITIONAL DESIGN CONSIDERATIONS:**
  - **Added:** Items 15–20. Item 15 notes the motor driver uses an IRFZ34N low-side MOSFET. Item 16 describes the flyback diode and decoupling capacitor. Item 17 describes the PWM channel on Timer2. Item 18 describes software debounce on the button. Item 19 describes the 6-state speed cycle. Item 20 notes the 9 V adapter supplies the motor.

- **SYSTEM DESIGN:**
    - **Added:** New subsection "Motor Driver" describing the low-side MOSFET driver circuit, Timer2 Fast PWM configuration, button debounce state machine, and integration with the cyclic executive as taskMotor.
- **TESTING AND VALIDATION REQUIREMENTS:**
    - **Added:** T.28, T.29, T.30, T.31, T.32, T.33, T.34, T.35, T.36.
- **TESTING RESULTS**
    - **Added:** Rows for T.28 – T.36.
    - **Changed:** GitHub repository link updated from Project 3 to Project 4.

Added Documents:

- ECN1.3 – This Engineering Change Note, dated 10/27/2026.
- motor.h – Motor driver header, CMPE311-PROJECT-ADDMOTOR-HDR.1.0, dated 10/27/2026.
- motor.c – Motor driver implementation, CMPE311-PROJECT-ADDMOTOR-SRC.1.0, dated 10/27/2026.

Removed Documents:

None.

Replaced Documents:

None.

Any Additional Design Considerations:

1. The motor is driven by an IRFZ34N N-channel MOSFET in a low-side switch configuration. PWM is applied to the gate through a $150\ \Omega$ resistor with a $10\ k\Omega$ pull-down.
2. A 1N4004 flyback diode protects the MOSFET from inductive spikes when the motor switches off. A $0.1\ \mu$F ceramic capacitor across the motor terminals suppresses brush noise.
3. PWM is generated by Timer2 in Fast PWM mode on Arduino pin 11 (OC2A), with no prescaler. Timer2 is independent of Timer0 (used by mymilli()) and does not affect the custom timer.

4. The pushbutton on Arduino pin 4 uses the internal pull-up and software debounce with a 30 ms threshold based on mymilli().
5. The fan cycles through six states: OFF, LOW, MEDIUM, HIGH, MEDIUM, LOW, then back to OFF. This implements the up-down cycle required by HL.15.
6. The motor is powered from the 9 V 1000 mA adapter via the Arduino Vin pin. The Arduino and motor share a common ground.
7. No third-party multitasking libraries or RTOS are used so EC.5 remains satisfied.
8. All Project 1, 2, and 3 customer requirements, high-level technical requirements, educational constraints, and user-visible behavior are unchanged.